
Chapter 2. Installing the Flex System Enterprise Chassis

Install the Flex System Enterprise Chassis in your facility by setting up and configuring all of the hardware components.

Before you begin the installation process, make sure that you have completed all planning activities. Planning information is available from <http://flexsystem.lenovofiles.com/help/index.jsp>.

Setting up your Flex System Enterprise Chassis

Follow the instructions in this section to set up the Flex System Enterprise Chassis hardware.

Before you begin, make the following preparations:

- Read “[Safety](#)” on page iii and “[Handling static-sensitive devices](#)” on page 958. This information will help you work safely.
- Confirm that each box contains all the components that are listed on the packing list. Your order might consist of more than one box, and each box might contain more than one component.
- Make sure that at least 10 units of contiguous space (10U) is available in the rack.

To set up the chassis hardware, complete the following tasks:

1. Remove the components from the chassis to decrease the weight so that it can be safely installed in the rack (see “[Removing components](#)” on page 27). You can also remove the chassis shelves to further reduce the weight. You do not have to remove the two fan logic modules in the rear of the chassis.
2. Install the chassis in a rack (see “[Installing the chassis in a rack](#)” on page 33).
3. Reinstall all of the components that you removed (see “[Installing components](#)” on page 41).
4. Cable the components in the chassis to the applicable external devices (see “[Cabling the chassis](#)” on page 54).
5. Connect the chassis to power (see “[Connecting the chassis to power](#)” on page 55).

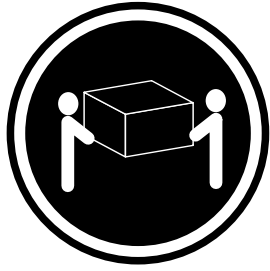
Note: The Flex System Enterprise Chassis does not have a power switch. To turn off chassis power, see “[Disconnecting the chassis from power](#)” on page 60.

Removing components

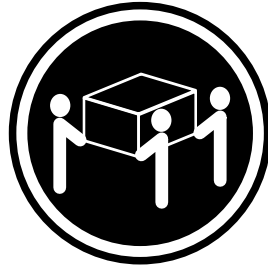
Remove components to reduce the weight of the Flex System Enterprise Chassis so that it can safely be installed in the rack.

Statement 4

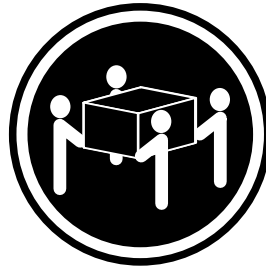




≥ 18 kg (39.7 lb)



≥ 32 kg (70.5 lb)



≥ 55 kg (121.2 lb)

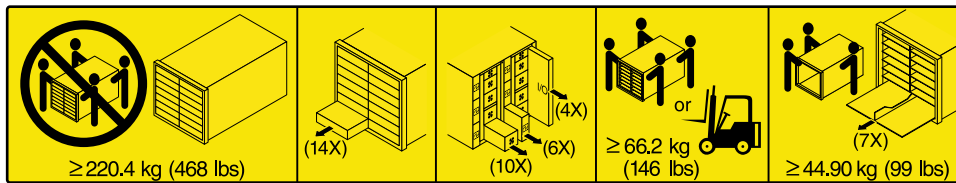
CAUTION:

Use safe practices when lifting.

Statement 32

CAUTION:

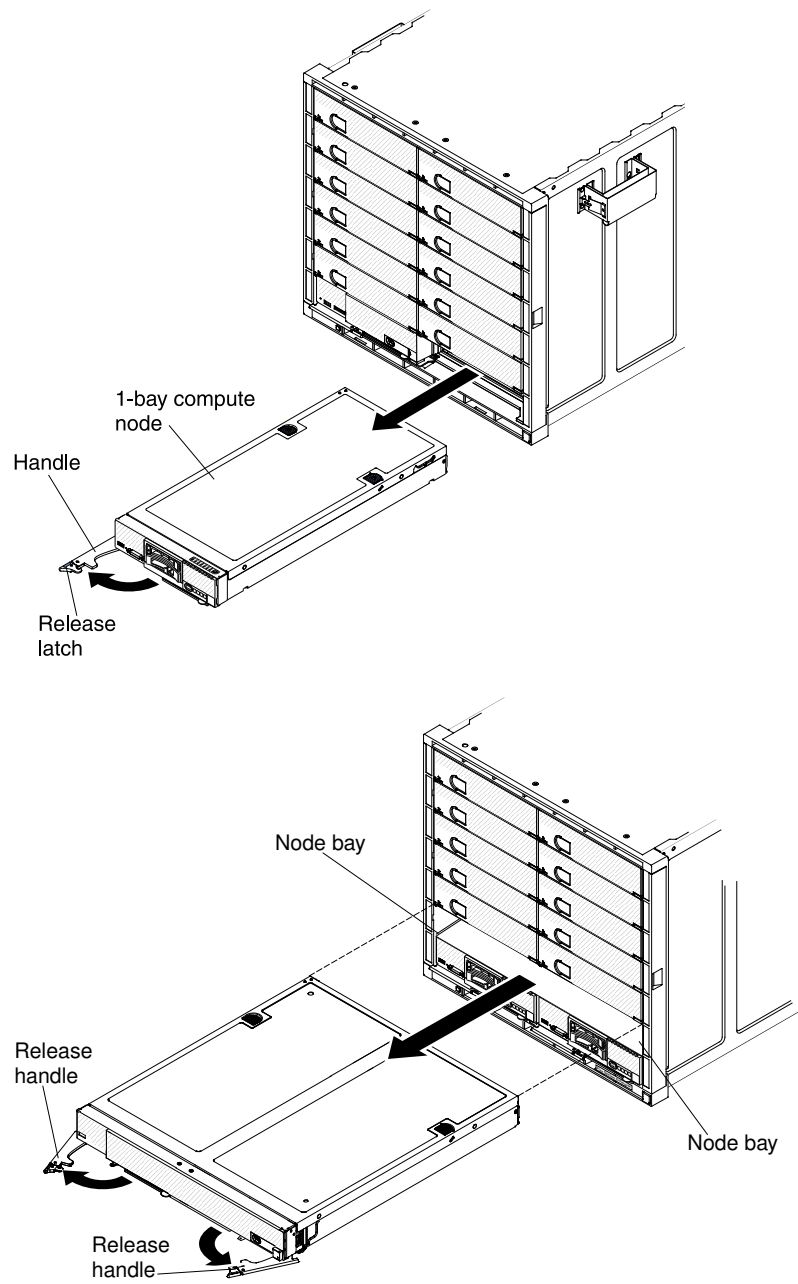
To avoid personal injury, before lifting the unit, remove all the blades, power supplies, and removable modules to reduce the weight.

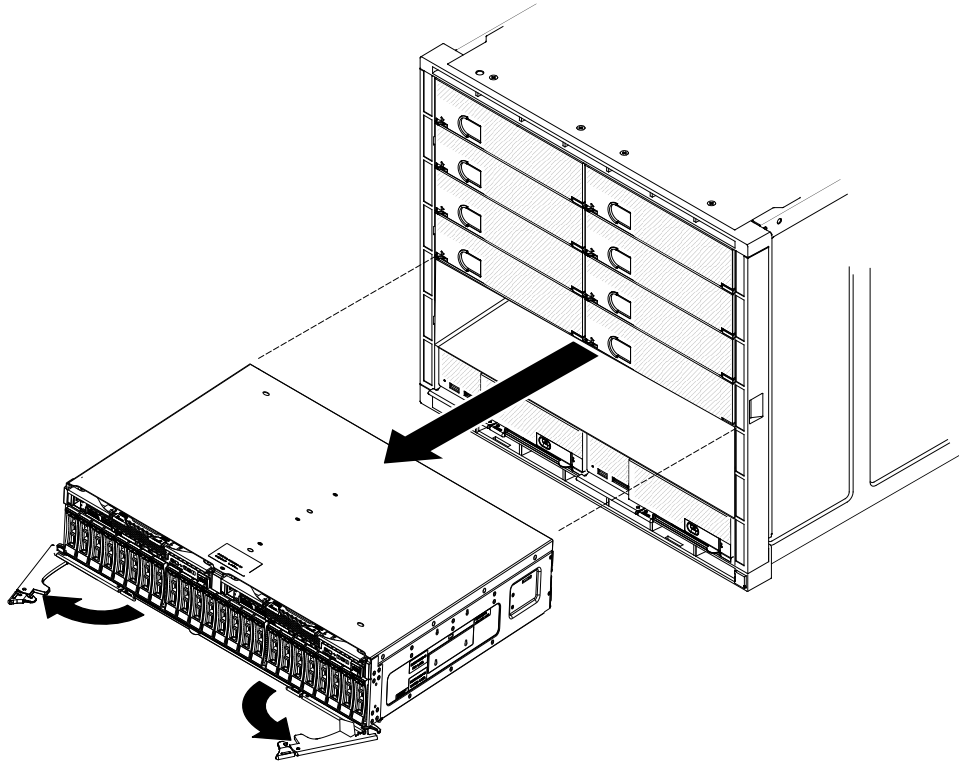


Complete the following steps:

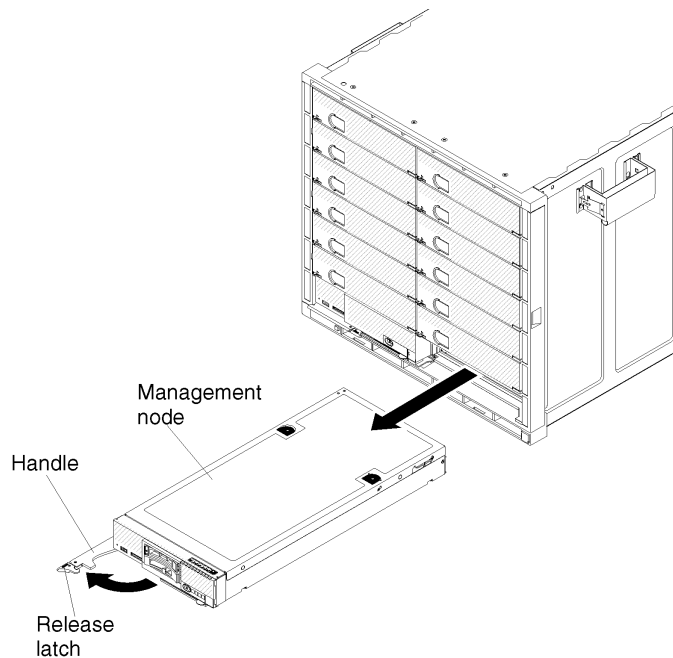
Step 1. Remove the components from the front of the chassis and place them on a flat, static-protective surface.

- a. Read [“Safety” on page iii](#) and [“Handling static-sensitive devices” on page 958](#).
- b. Remove all of the installed compute nodes or storage nodes.





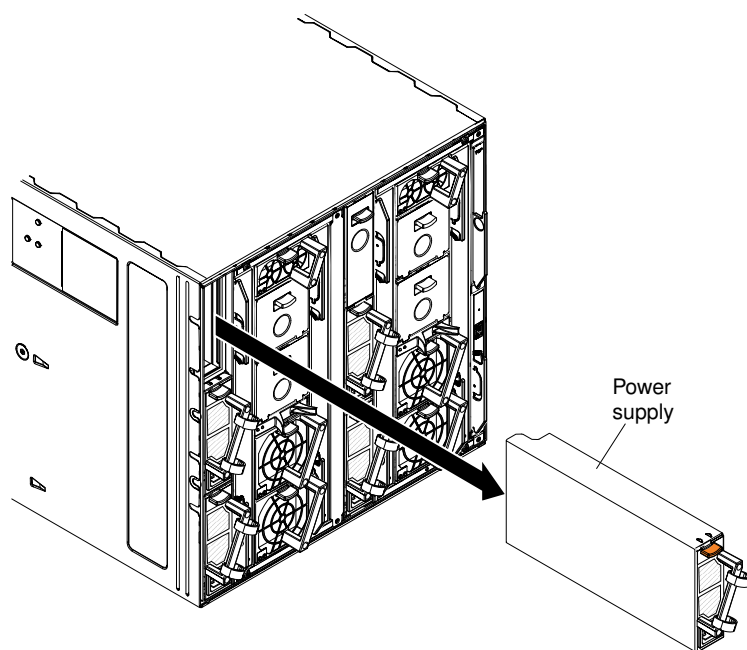
- c. Remove the Flex System Manager management node, if one is installed.



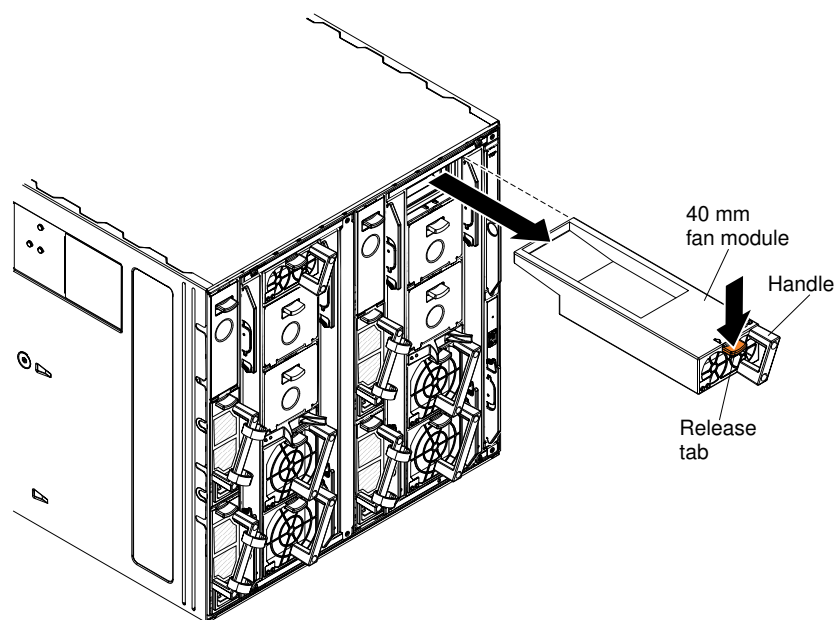
- d. Remove the node bay fillers and the chassis shelves, if you want to reduce the weight of the chassis further. See [“Removing a chassis shelf” on page 985](#) for instructions.

Step 2. Remove the components from the rear of the chassis and place them on a flat, static-protective surface.

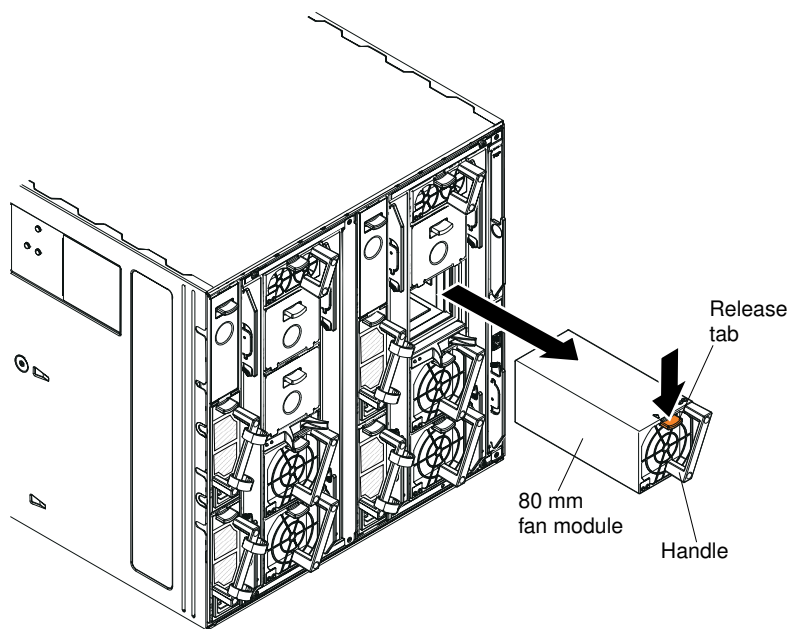
- a. Remove all of the power supplies.



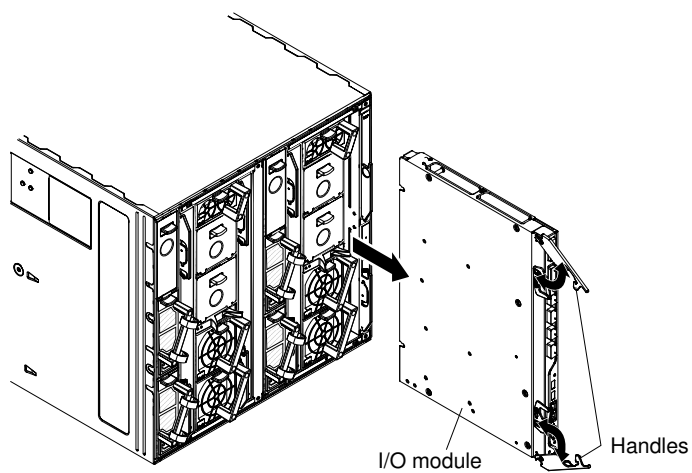
- b. Remove both of the 40 mm fan modules.



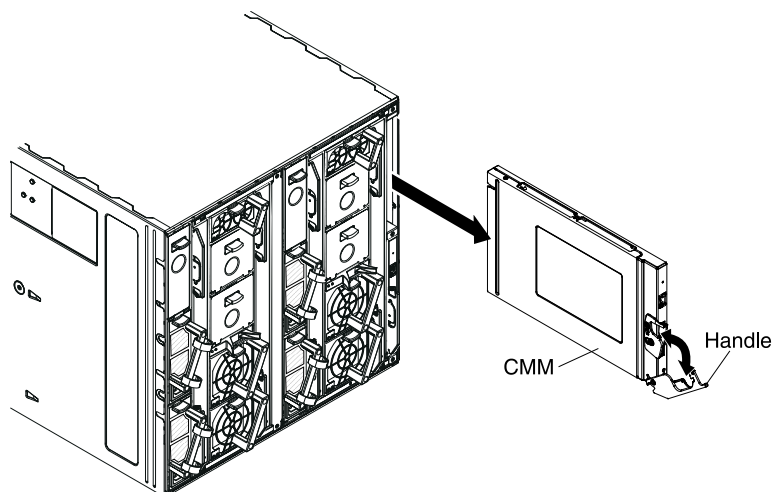
- c. Remove all of the 80 mm fan modules.



d. Remove all of the I/O modules.



e. Remove all of the Chassis Management Modules.



Installing the chassis in a rack

Use the information in this section, the rack template, and the rack installation kit that comes with the Flex System Enterprise Chassis to install it in a rack.

Rack Safety Information, Statement 2



DANGER

- **Always lower the leveling pads on the rack cabinet.**
- **Always install stabilizer brackets on the rack cabinet.**
- **Always install servers and optional devices starting from the bottom of the rack cabinet.**
- **Always install the heaviest devices in the bottom of the rack cabinet.**

Before you begin, review the [“Installation guidelines” on page 957](#).

Rack requirements:

- Make sure that the room air temperature is below 40°C (104°F).
- Do not block any air vents; usually, 15 cm (6 in.) of air space in the rear and 5 cm (2 in.) in the front provides proper airflow.
- Three or more people are required to install the device in a rack.
- Do not leave any unused space within a rack open. Fillers must be used to prevent recirculation of warm air.
- Install your Flex System Enterprise Chassis only in a rack that has perforated front and rear doors or in a rack that is equipped with a Rear Door Heat eXchanger.
- Do not extend more than one device out of the rack at the same time.
- Remove the rack doors and side panels to provide easier access during installation.
- The EIA flanges must have holes and clearances per EIA-310-D.
- If you have an adjustable rack, set the distance between the front and rear EIA flanges to 719 mm (28.3 inches) outside to outside.
- Make sure that there is sufficient room in front of the front EIA flange to provide minimum bezel clearance of 50 mm (1.97 inches).
- Make sure that there is sufficient room behind the rear of the rear EIA flanges to provide for cable management and routing.
- Leave 1U of empty space at the top of the rack, if the cables exit at the top, or leave 1U of empty space at the bottom of the rack, if the cables exit at the bottom. This will ensure that the cables don't block service access to replaceable components.
- Rack weight-handling capacity must be sufficient for the aggregate weight of the populated chassis, power distribution units, and power cables.
- The rack must be stabilized with stabilization brackets and leveling pads so that it does not become unstable when it is fully populated.

To install the chassis in a rack, complete the following steps:

Step 1. Read [“Safety” on page iii](#) and [“Handling static-sensitive devices” on page 958](#).

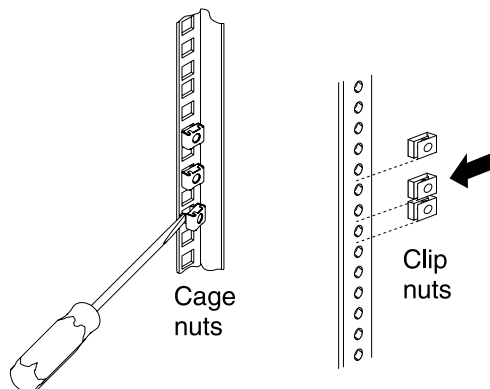
Step 2. Open the rail kit and the rack installation kit that come with the chassis:

- The rail kit box includes the following hardware:
 - One right chassis support rail
 - One left chassis support rail
 - Eight M5x16 combi-head screws (black)
 - Ten strain relief ties
- The rack installation kit includes the following hardware:
 - Twelve M5 cage nuts (use on EIA flanges with square holes)
 - Twelve M5 clip nuts (use on EIA flanges with round holes)
 - Ten M5x16 combi-head screws (black)
 - Eight M5x16 captive washer screws (silver)
 - One right rear support bracket
 - One left rear support bracket
 - One lower rear support bracket

Step 3. If the rack has a door, remove it.

Step 4. Position the rack template that comes with the chassis on the front of the rack at the location where you want to install the chassis. Make sure that the template does not overlap any installed devices, and align the template with the holes in the rack (for an illustration of the template see [“Rack template” on page 39](#)). For EIA flanges with square holes, install M5 cage nuts from the rack installation kit in the holes that are indicated on the template.

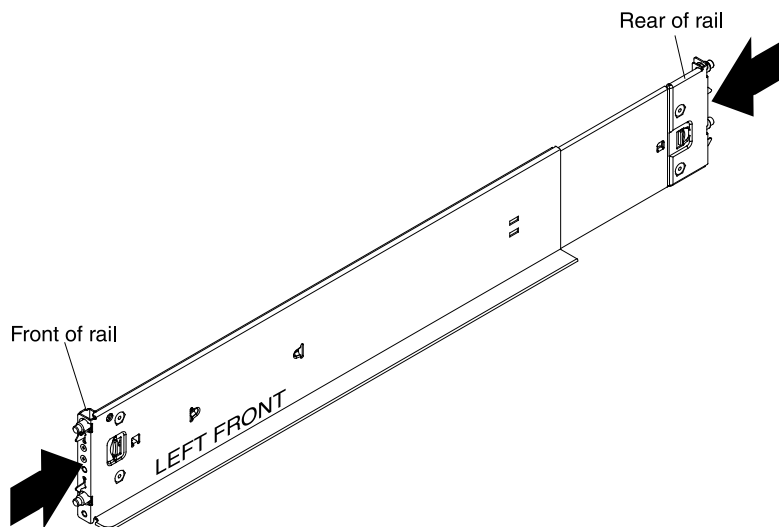
Note: If the EIA flange has round holes, install the M5 clip nuts from the installation kit instead of the M5 cage nuts.



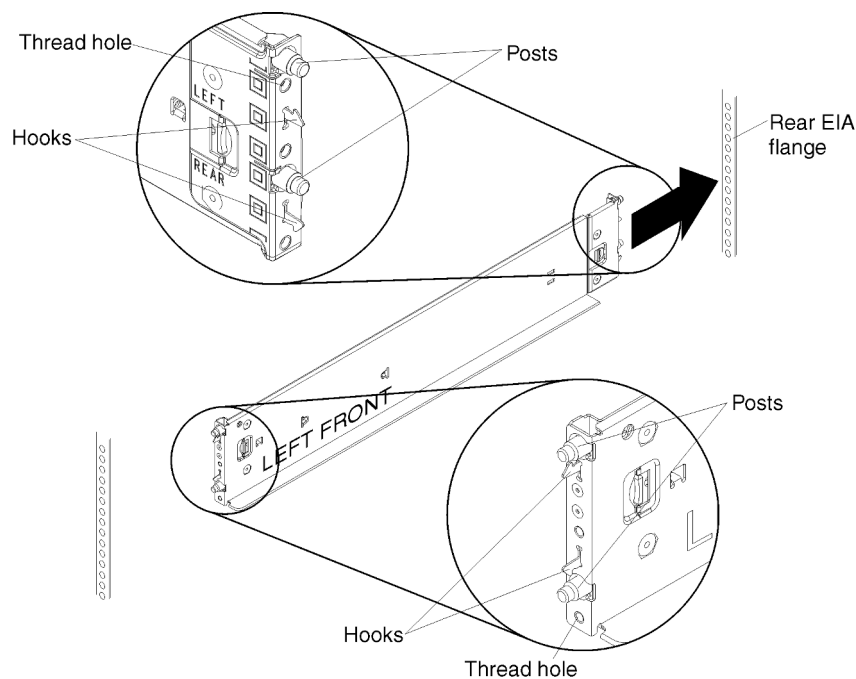
Step 5. Position the rack template on the back of the rack, and install M5 cage nuts in the square holes that are indicated on the template. If the EIA flange has round holes, install the M5 clip nuts from the installation kit.

Note: Make sure that a cage nut or clip nut is installed in every hole indicated on the template.

Step 6. Retract both chassis rails, if they are not already retracted. Rail posts and locking hooks are on each end of each rail.



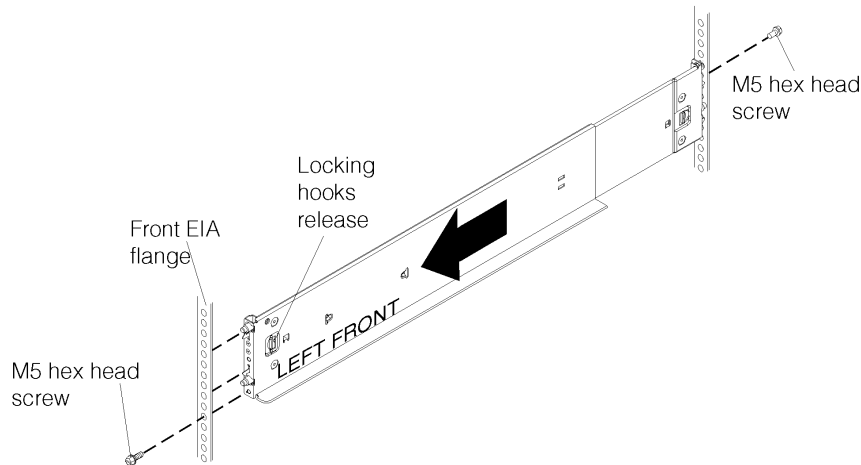
- Step 7. Position the left chassis rail in the selected location on the rear of the rack and align the posts on the chassis rail with the holes on the back EIA flange. Insert the posts on the rear of the chassis rail through the holes on the rear EIA flange until the hooks snap into place.



Note:

- Be sure to align the bottom edge of the chassis rail with the bottom U that you want the chassis to rest on.
- If you misalign the chassis rail when you insert it into the holes in the EIA flange, press the locking hooks release on the chassis rail to disengage the hooks and slide the posts out of the holes on the EIA flange. Reinsert the rail into the correct holes on the EIA flange.
- Make sure that the rail posts protrude through the holes on the EIA flanges.

- Step 8. Pull the chassis rail forward and insert the posts on the front of the rail into the corresponding holes on the front EIA flange until it snaps into place. Repeat steps 6 through 8 for the right chassis rail.

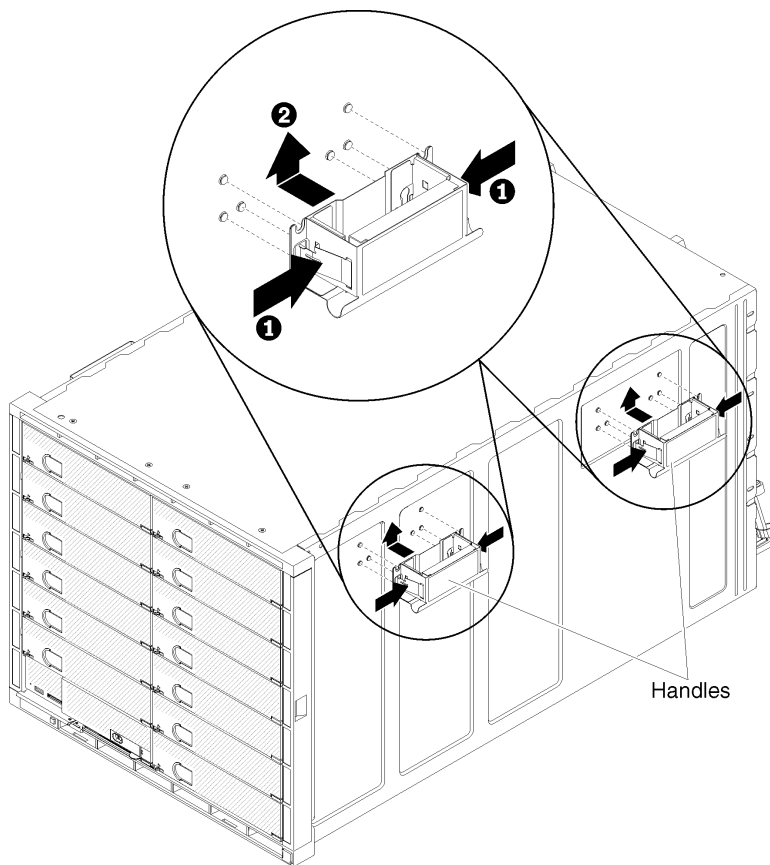


Step 9. Install M5x16 combi-head screws (black) from the rail kit to secure the rails to the EIA flange:

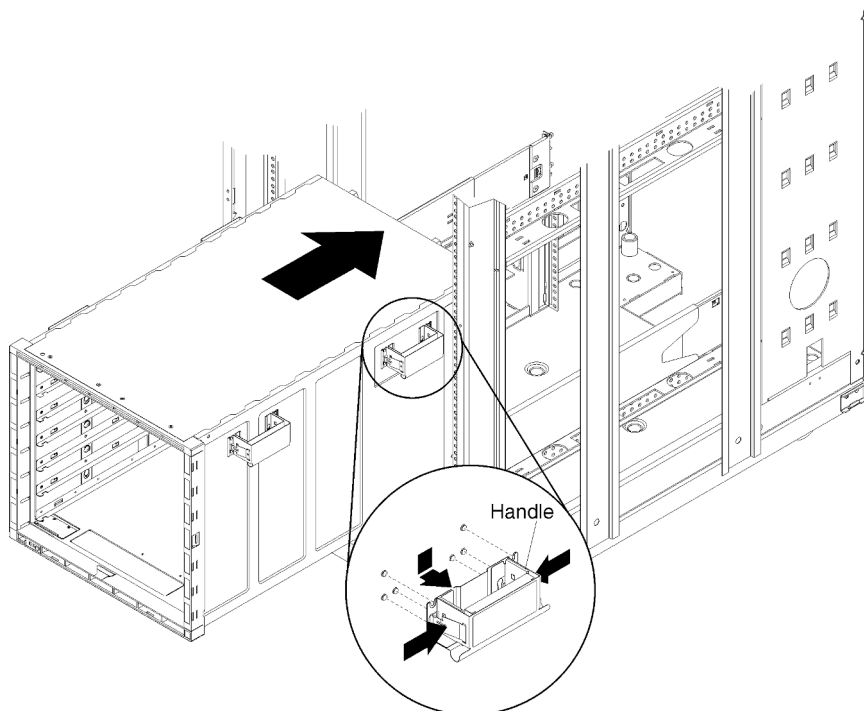
1. At the front of the rack, install an M5x16 combi-head screw (black) in the lower threaded hole on the front of each chassis rail (two screws required, one in the right rail and one in the left rail).
2. At the rear of the rack, install an M5x16 combi-head screw (black) in the upper threaded hole on the rear of each chassis rail (two screws required, one in the right rail and one in the left rail).
3. Tighten the front and rear rail screws to 30 in-lbs (3.4 Nm).

Step 10. Remove the components from the chassis, to make the chassis easier to install in the rack. See [“Removing components” on page 27](#) for instructions. You do not have to remove the two fan logic modules in the rear of the chassis. You can also remove the chassis shelves to further reduce the weight.

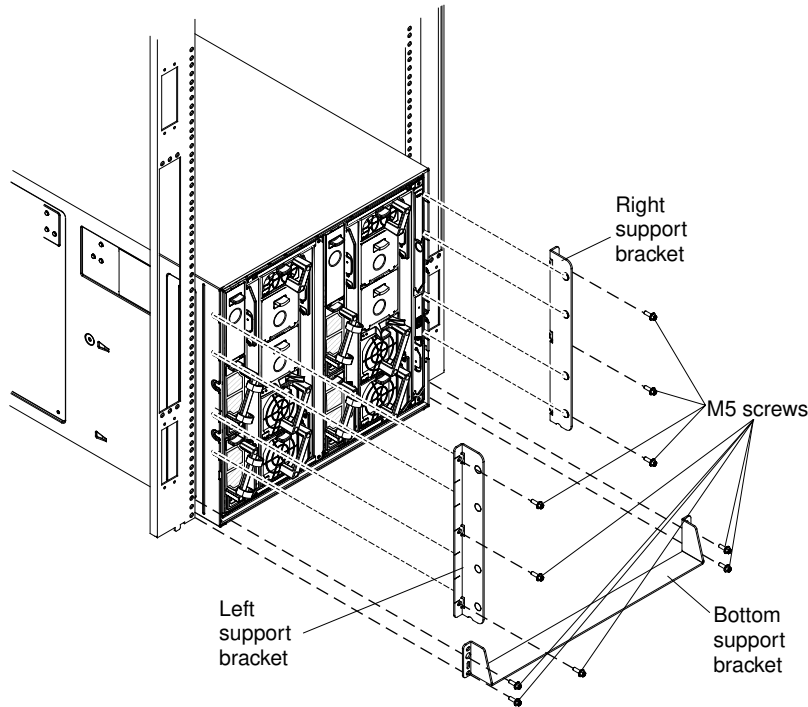
Step 11. Attach the chassis handles to the chassis. Four handles come in a box with the chassis. Align the slots on each handle with the posts on the side of the chassis and slide the handle up until it locks into place.



Step 12. Lift the chassis up, place the rear of the chassis onto the chassis rails, and slide the chassis into the rack until the rear chassis handles are near the front EIA flanges. While you support the chassis, remove each rear handle from the chassis (both sides) by pressing inward on the spring-latches on the handle and sliding the handles down to remove them.

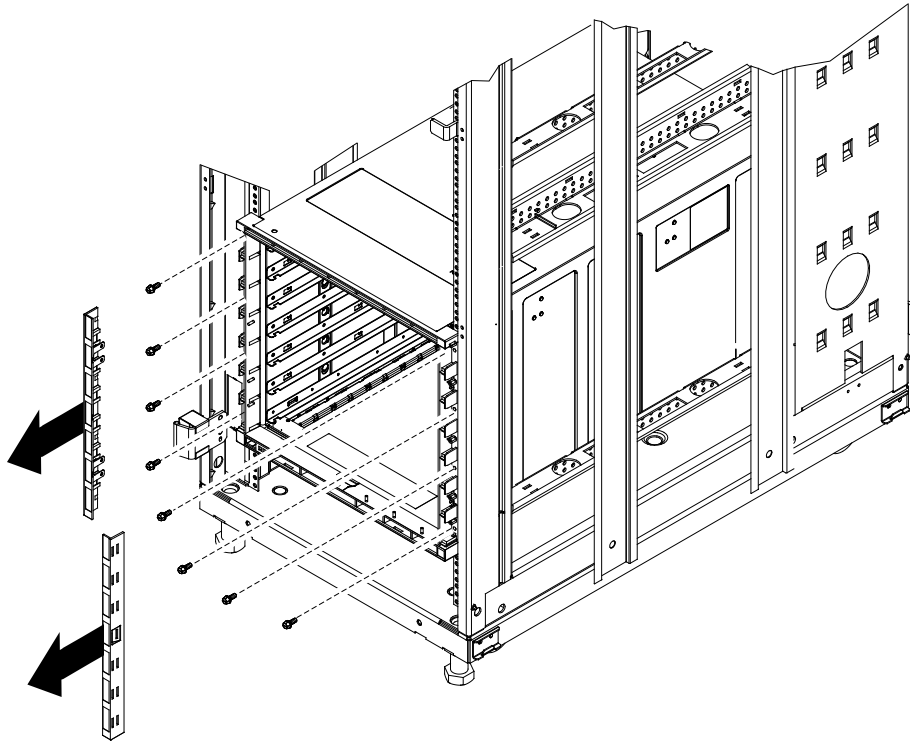


- Step 13. Slide the chassis farther into the rack until the front chassis handles are near the front EIA flanges, and remove the front handles. Then, slide the chassis all the way into the rack.
- Step 14. If you plan to transport the rack to another location with the chassis installed, you must install the rear support brackets. Rear support brackets and mounting screws are provided in the rack installation kit.



- Facing the rear of the chassis, align the left support bracket with the four slots on the outside of the chassis.
- Slide the support bracket forward until it locks into place in the slots.
- Install three M5x16 combi-head screws (black) from the rack installation kit to secure the support bracket to the rack, but do not tighten the screws.
- Repeat steps 15a through 15c for the right support bracket.
- Fit the lower support bracket to the bottom of the chassis; then, slide the support bracket forward against the rack.
- Install four M5x16 combi-head screws (black) from the rack installation kit in the lower support bracket, two screws on the right side and two screws on the left side, but do not tighten the screws.
- After the right, left, and lower support brackets and screws are installed, tighten all of the screws to 30 in-lbs (3.4 Nm).

- Step 15. Remove the plastic label plates from the left and right sides of the chassis.



Step 16. Install eight M5x16 captive washer screws (silver) from the rack installation kit through the front of the chassis (four screws on each side), but do not tighten the screws.

Step 17. After all of the screws are installed, tighten the screws to 30 in-lbs (3.4 Nm).

Step 18. Reinstall the plastic label plates onto the left and right sides of the chassis.

Step 19. Reinstall the chassis shelves, if you removed them earlier.

Step 20. Reinstall the components (see [“Installing components” on page 41](#)).

Rack template

The rack template comes with the Flex System Enterprise Chassis.

If you do not have the rack template, you can use the template illustration as a guideline to identify the mounting holes on the front and rear of the rack.

Note: Use this rack template as a guideline only. If you print the template, it might not be printed to scale. You can download a full-size template from http://flexsystem.lenovofiles.com/help/topic/com.lenovo.acc.8721.doc/printable_doc.html

Installing components

After you install the Flex System Enterprise Chassis in a rack, install all of the components in the chassis.

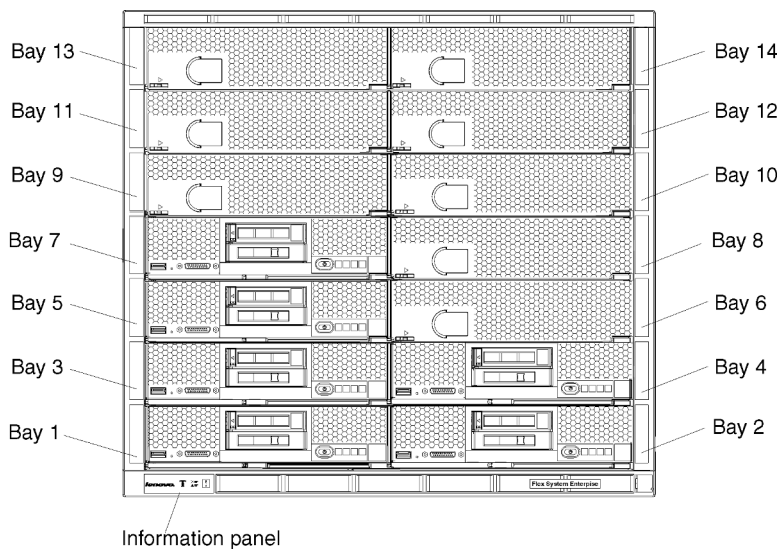
Before you install components in the Flex System Enterprise Chassis:

- Read “Safety” on page iii and “Installation guidelines” on page 957.
- Make sure that the Chassis Management Module firmware is the latest level available. See [Updating the CMM firmware](#) “Updating the CMM firmware” in the *Lenovo Flex System Chassis Management Module Installation Guide* for more information.

Important: When you install compute nodes, power supplies, and 80 mm fan modules in the Flex System Enterprise Chassis, install them starting from the bottom up.

Compute nodes

The following illustration shows the locations of node bays in the Flex System Enterprise Chassis.



Note:

A 1-bay compute node occupies one node bay in the chassis.

A 1-bay compute node with captive-mode expansion node occupies two adjacent node bays (horizontally) in the chassis and requires the chassis shelf to be removed prior to installation.

A 2-bay compute node occupies two adjacent node bays (horizontally) in the chassis and requires the chassis shelf to be removed prior to installation.

A 4-bay storage node or storage expansion enclosure occupies four adjacent node bays (two vertically and horizontally) in the chassis and requires the chassis shelves and shelf supports to be removed prior to installation.

Power supplies

The chassis power load is dependent on the number and type of compute nodes, storage nodes, I/O modules, and management modules that are installed in the chassis. The number and type of power supplies that you install determines how much power is available to power all of these devices. The amount of power available must be sufficient for the expected chassis load.

You can use the Lenovo Capacity Planner to determine the power load for a specific chassis configuration. See <https://datacentersupport.lenovo.com/tw/en/products/solutions-and-software/software/lenovo-capacity-planner/solutions/ht504651> for more information.

Important:

- Up to six power supplies can be installed.
- Do not mix different types of power supplies in the Flex System Enterprise Chassis.
- Each chassis must contain either all ac-powered supplies or all dc-powered supplies.
- For ac-powered chassis, do not mix 2100 W and 2500 W power supplies. Chassis powered by ac power should contain only power supplies of the same wattage.
- Install power supplies from the bottom up starting with power bays 1 and 4, then power bays 2 and 5, then power bays 3 and 6.

Install power supplies according to the number of populated node bays as shown in [Table 6 “Required power supplies relative to the number of populated node bays” on page 42](#). See also “Power supplies” on [page 20](#) for more information.

Table 6. Required power supplies relative to the number of populated node bays

Number of populated node bays	Power supplies required
1 - 4	2
5 - 6	3
7 - 8	4
9 - 11	5
12 - 14	6

80 mm fan modules

The number and type of 80 mm fans that you install depends on the type of power supplies that are installed in the chassis and the number of populated node bays. Up to eight 80 mm fan modules can be installed.

Important:

- Chassis with 2100 W power supplies must have 80 mm single-fan modules.
- Chassis with 2500 W power supplies must have 80 mm dual-fan modules.
- Use only the same type of 80 mm fan module in each chassis. Do not mix single-fan and dual-fan 80 mm fan modules within a chassis.

Install 80 mm fan modules according to the number of populated node bays as shown in [Table 7 “Required 80 mm fan modules relative to the number of populated node bays” on page 42](#). See also “Fan modules” on [page 22](#) for more information.

Table 7. Required 80 mm fan modules relative to the number of populated node bays

Fan zone 1 (behind node bays 1, 3, 5, 7, 9, 11, and 13)		Fan zone 2 (behind node bays 2, 4, 6, 8, 10, 12, and 14)	
Number of populated node bays	80 mm fans required	Number of populated node bays	80 mm fans required
1 - 2	2	1 - 2	2
3 - 4	3	3 - 4	3
5 - 7	4	5 - 7	4

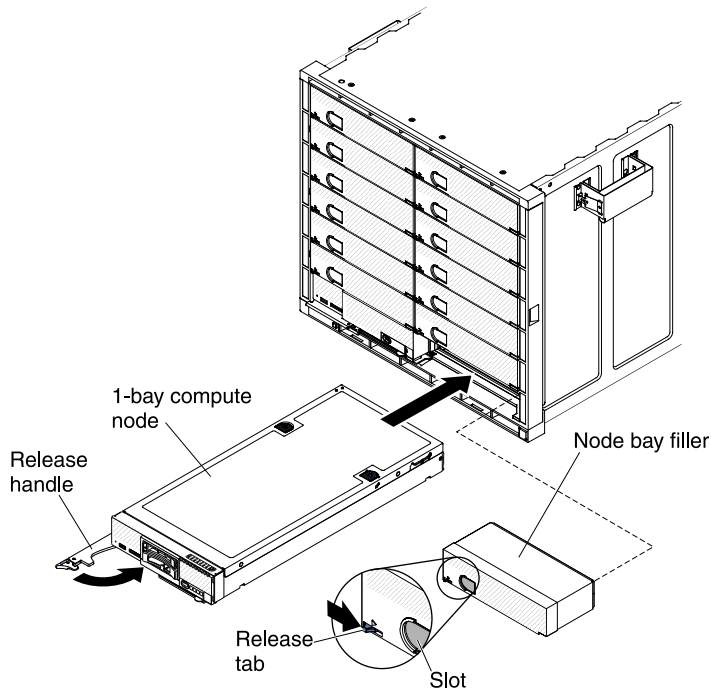
Installing a 1-bay compute node

You can install up to fourteen 1-bay compute nodes in the Flex System Enterprise Chassis. Compute nodes are installed in the front of the Flex System Enterprise Chassis.

Before you install a compute node in the Flex System Enterprise Chassis, complete the following steps:

1. Verify that the compute node is compatible with the chassis. See <http://www.lenovo.com/serverproven/>.
2. Make sure that enough power supplies and fan modules are installed in the chassis to support the compute node. See “Installing components” on page 41 to determine the number of power supplies and 80 mm fan modules that are required and where they should be installed in your chassis configuration.
3. Read the instructions that come with the compute node.
4. Make sure that you have installed any optional hardware devices in the compute node.
5. Select the bay for the compute node.

To install a 1-bay compute node, complete the following steps.



- Step 1. Remove the node bay filler, if one is installed. Push the filler release tab to the right; then, grasp the filler by the slot and pull it out of the bay.
- Step 2. Open the release handle (rotate the handle to the left).
- Step 3. Slide the compute node into the node bay until it is seated.
- Step 4. Close the release handle (rotate the handle to the right).

After you install the compute node, make a note of the compute node identification information on one of the labels that come with the Flex System Enterprise Chassis. Place a label on the node label tab and on the adjacent chassis label plate, to the right or left of the compute node (depending on the bay in which the compute node is installed). See “User labels” on page 59 for more information.

Important: Do not place the label on the compute node or in any way block the ventilation holes on the chassis.

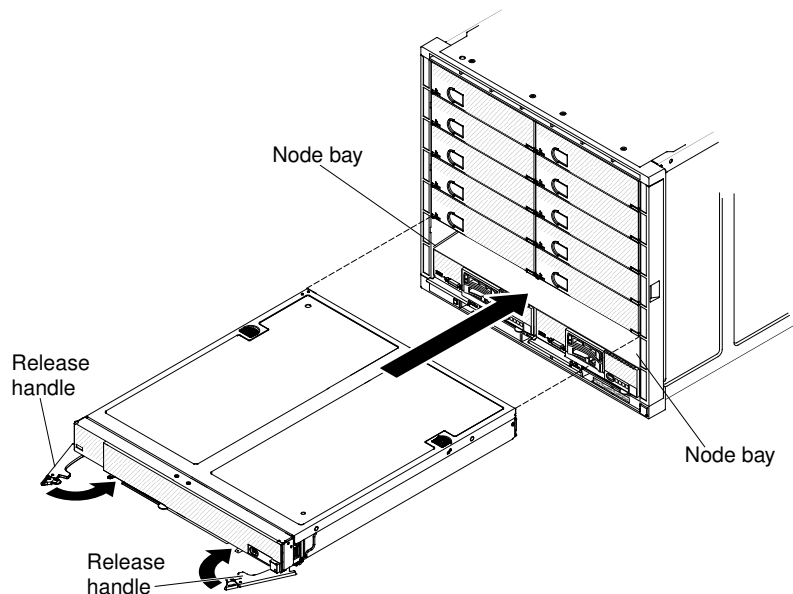
Installing a 2-bay compute node

You can install up to seven 2-bay compute nodes in the chassis. A 2-bay device occupies two adjacent node bays (horizontally) in the chassis.

Before you install a compute node in the Flex System Enterprise Chassis, complete the following steps:

1. Verify that the compute node is compatible with the chassis. See <http://www.lenovo.com/serverproven/>.
2. Make sure that enough power supplies and fan modules are installed in the chassis to support the compute node. See “Installing components” on page 41 to determine the number of power supplies and 80 mm fan modules that are required and where they should be installed in your chassis configuration.
3. Read the instructions that come with the compute node.
4. Make sure that you have installed any optional hardware devices in the compute node.
5. Select the bays for the compute node. Two adjacent bays are required.
6. Remove the fillers and the chassis shelf from the selected bays, if they are installed. See “Removing a chassis shelf” on page 985 for instructions.

To install a 2-bay compute node, complete the following steps.



- Step 1. Open both release handles (rotate the left handle to the left and rotate the right handle to the right).
- Step 2. Slide the compute node into the node bays until it is seated.
- Step 3. Close both release handles.

After you install the compute node, make a note of the compute node identification information on one of the labels that come with the Flex System Enterprise Chassis. Place a label on the node label tab and on the adjacent chassis label plate, to the left of the compute node. See “User labels” on page 59 for more information.

Important: Do not place the label on the compute node or in any way block the ventilation holes on the chassis.

Installing a 4-bay storage enclosure

A 4-bay storage enclosure occupies four adjacent node bays in the chassis.

A storage system includes a control enclosure and might also include one or more expansion enclosures. Refer to the documentation that comes with the storage enclosure for detailed information about installing, cabling, and configuring the storage enclosure. Storage node documentation is available from http://flexsystem.lenovofiles.com/help/topic/com.lenovo.acc.4939.doc/site_product_page.html.

Before you install a 4-bay storage enclosure in the Flex System Enterprise Chassis, complete the following steps:

1. Verify that the storage enclosure is compatible with the chassis. See <http://www.lenovo.com/serverproven/>.
2. Make sure that enough power supplies and fan modules are installed in the chassis to support the storage enclosure and any expansion enclosures that you plan to install. See “Installing components” on page 41 to determine the number of power supplies and 80 mm fan modules that are required and where they should be installed in your chassis configuration.
3. Select the bays for the storage enclosure. Four adjacent bays (two vertically and horizontally) are required.
4. Remove the node fillers and the chassis shelves from the selected bays, if they are installed. See “Removing a chassis shelf” on page 985 for instructions.
5. Remove the shelf supports from the center of the selected bays, if they are installed. See “Removing the shelf supports” on page 960 for instructions.

Complete the following steps:

- Step 1. Follow the detailed instructions that come with the 4-bay storage enclosure to install, cable, and configure the storage enclosure. Storage node documentation is available from http://flexsystem.lenovofiles.com/help/topic/com.lenovo.acc.4939.doc/site_product_page.html.

After you install the storage enclosure, complete the following steps:

1. Make a note of the enclosure identification information on one of the labels that come with the Flex System Enterprise Chassis. Place a label on the enclosure label tab and on the adjacent chassis label plate, to the left of the storage enclosure. See “User labels” on page 59 for more information.

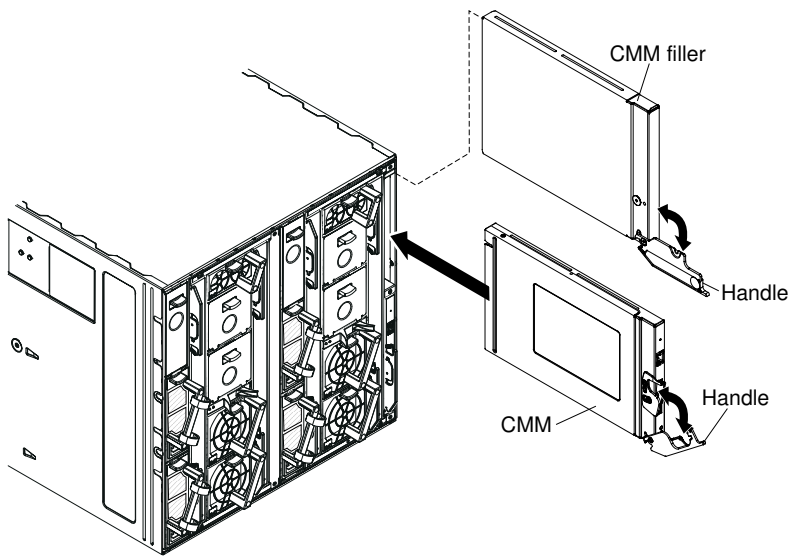
Important: Do not place the label on the storage enclosure or in any way block the ventilation holes on the chassis.

Installing a Chassis Management Module

You can install up to two Chassis Management Modules for redundancy support in the Flex System Enterprise Chassis.

Before you install the Chassis Management Module, read the installation instructions that come with the Chassis Management Module (see http://flexsystem.lenovofiles.com/help/topic/com.lenovo.acc.cmm.doc/cmm_product_page.html).

To install a Chassis Management Module (CMM), complete the following steps.



- Step 1. If a filler is installed in the CMM bay, remove it. Rotate the release handle on the filler down and slide it out of the bay.
- Step 2. Press the CMM release latch down and rotate the CMM handle down until it stops.
- Step 3. Align the CMM with the bay and slide it into the bay until it is seated.
- Step 4. Close the handle (rotate the handle up) so that it locks in place.

Note: Make sure that the power-on LED on the CMM is lit. This indicates that the CMM is operating correctly. See [“CMM controls and indicators” on page 17](#) to locate the LED.

When you install a CMM, if the chassis is not connected to a DHCP server on the network, it takes up to 3 minutes for the CMM to use the default (static) IP address.

After failover, you might have to wait as long as 5 minutes to establish a network connection to the CMM. Some networks include switches, routers, and hubs that do not allow (or relay) an address resolution protocol (ARP) from the new CMM to update the network cached ARP table. Without this information relay, the new MAC address/IP association will not recognize the CMM. This condition will correct itself after the ARP table times out. To prevent this condition, reconfigure the network-routing setup tables to enable ARPs to be relayed from the CMM.

After you install the Chassis Management Module, complete the following steps:

1. Connect all cables to the CMM.
2. Configure the CMM, by loading a previously saved configuration or by going through the Flex System Manager management software configuration wizard.

Installing a Flex System Manager management node

You can install one Flex System Manager management node in a multiple-chassis configuration. You can install up to two Flex System Manager management nodes in a multiple-chassis configuration for redundancy. One management node can be installed in two of the chassis in a multiple-chassis configuration.

Before you install the Flex System Manager management node, complete the following steps:

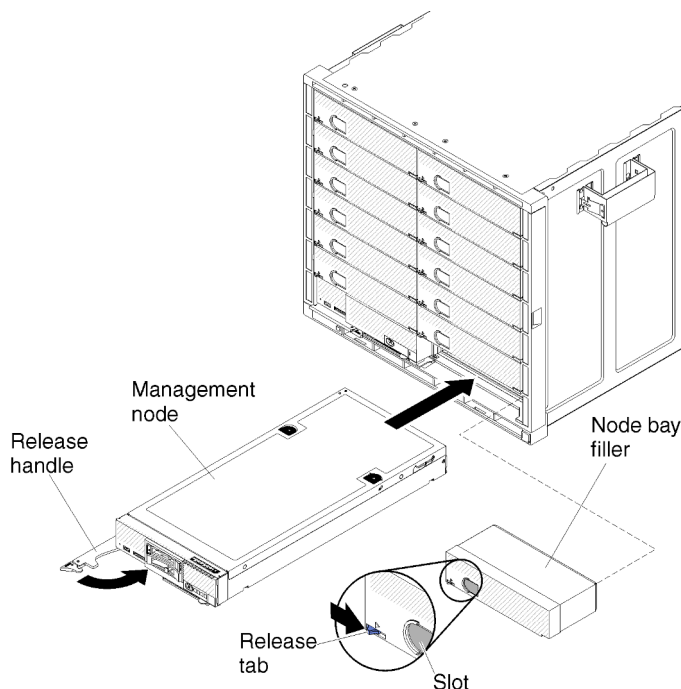
1. Make sure that enough power supplies and fan modules are installed in the chassis to support the management node. See [“Installing components” on page 41](#) to determine the number of power supplies and 80 mm fan modules that are required and where they should be installed in your chassis configuration.

2. Read the Flex System Manager management node hardware description (see the *Flex System Manager Installation and Service Guide* for more information).
3. Select the bay for the management node.

Attention:

- If you are installing two management nodes in separate chassis of a multiple-chassis configuration, they must *not* be installed in the same node bay. For example, if you install the first management node in node bay 1 in one of the chassis, install the second management node in a *different* chassis in any node bay other than node bay 1.
- The management nodes default static IP address depends in part on the node bay that it is installed in. If you install two management nodes in the same node bay in different chassis in a multiple-chassis configuration, they will have the same default IP address.

To install a Flex System Manager management node, complete the following steps.



- Step 1. Remove the bay filler, if one is installed. Push the filler release tab to the right; then, grasp the filler by the slot and pull it out of the bay.
- Step 2. Open the release handle (rotate the handle to the left).
- Step 3. Slide the management node into the chassis bay until it is seated.
- Step 4. Close the release handle (rotate the handle to the right).

After you install the Flex System Manager management node, complete the following steps:

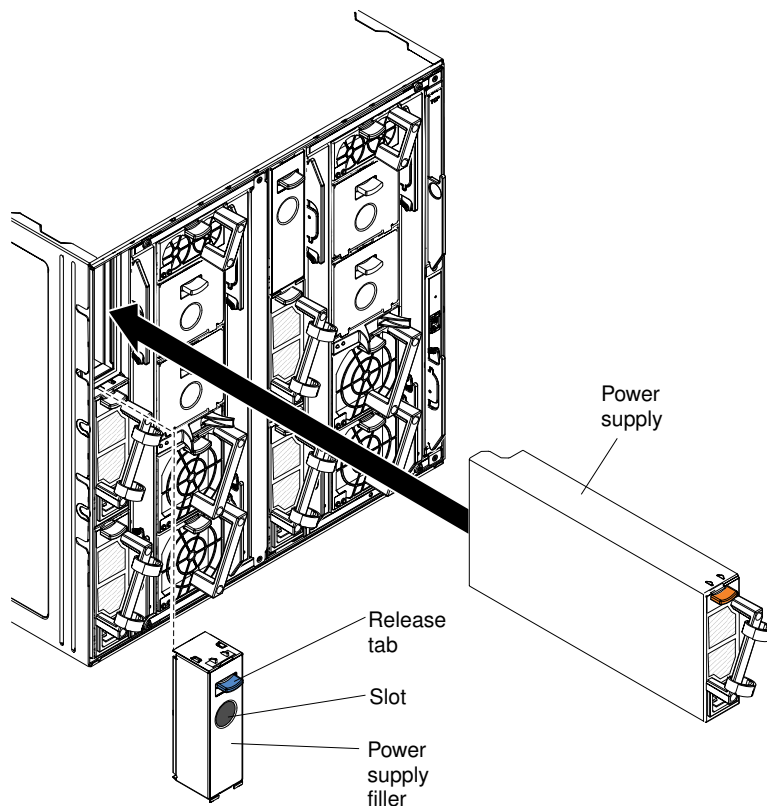
1. Connect all cables to the management node.
2. Configure the management node (see <http://flexsystem.lenovofiles.com/help/index.jsp> for more information).

Installing a power supply

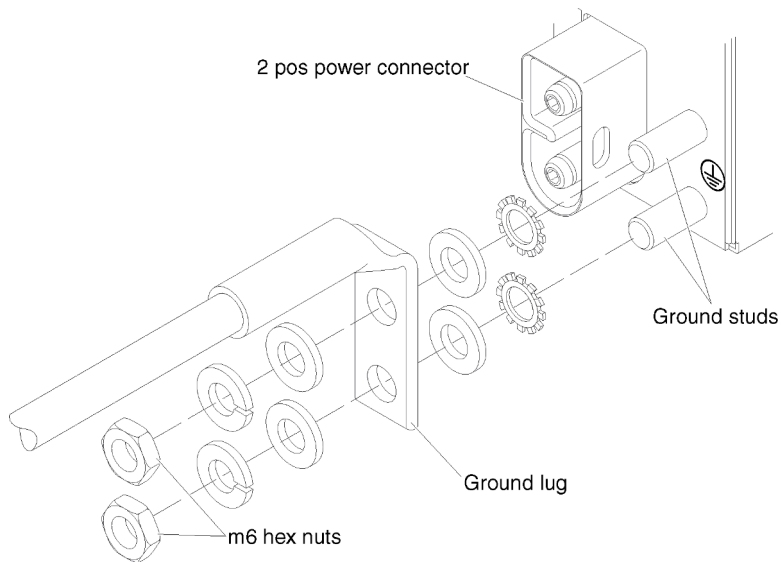
The Flex System Enterprise Chassis comes with two power supplies already installed in the rear of the chassis. You can install up to four additional power supplies in the chassis for a total of six power supplies.

Note: The number of power supplies that you install is dependent on the type of power supply, the chassis power load, and selected chassis power policy.

To install a power supply, complete the following steps.

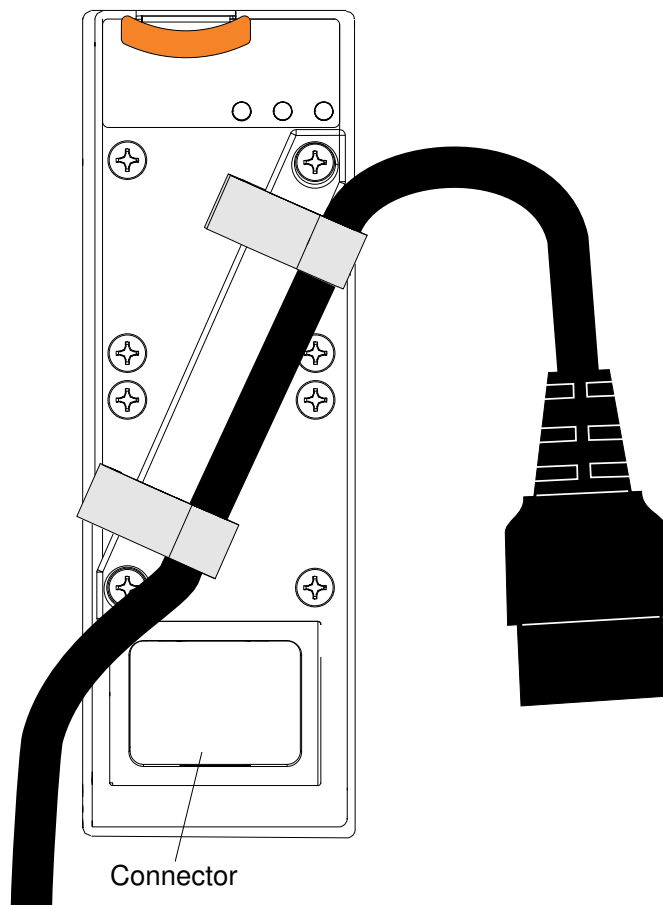


- Step 1. If you are adding a power supply, remove the filler from the power-supply bay in which you want to install the power supply (press the release tab, grasp the filler by the slot, and pull it out of the bay).
- Step 2. Grasp the power-supply handle and slide the power supply into the bay until it locks in place.
- Step 3. If you are installing a -48 to -60 V dc power supply, connect the earth ground cable to the power supply.
1. Use a 10 mm nut driver to remove the hex nuts from the ground studs.
 2. Remove the lock washer and one of the flat washers from each ground stud.
 3. Push the ground lug onto the ground studs; then, place the flat washer, the lock washer, and the hex nut back on each ground stud.
 4. Use a 10 mm nut driver to tighten the hex nuts to 4.0 - 4.8 Newton-meters (35.4 - 42.5 inch-pounds).

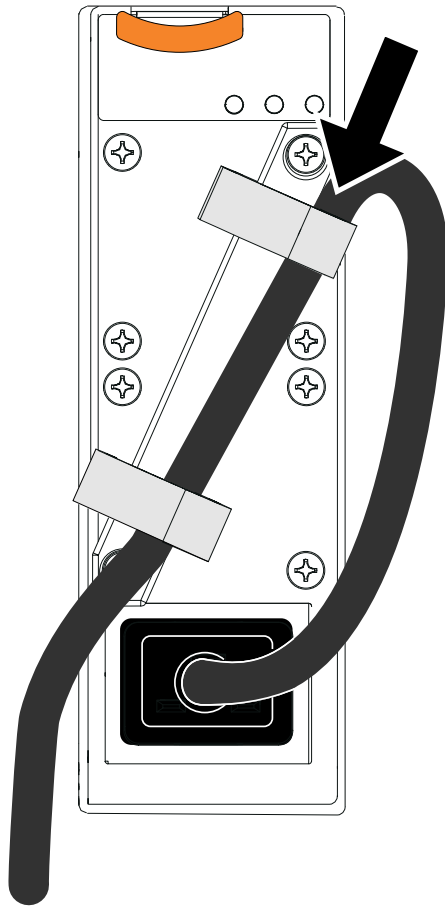


Step 4. Connect the power cord to the power supply:

1. Loosen the strain-relief ties that are attached to the power-supply handle, but do not remove them.
2. Align the power cord with the power-supply handle; then, secure the cord to the handle with the strain-relief ties.



3. Loop the power cord connector around and connect it to the power supply.



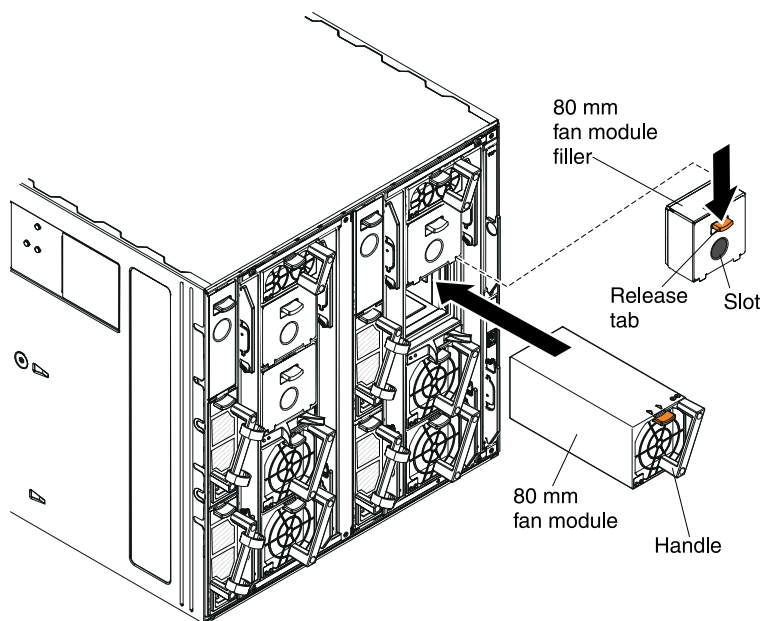
4. Push the power cord back through the strain-relief ties to remove excess cable from the loop.

Installing a 80 mm fan

The Flex System Enterprise Chassis comes with four 80 mm fan modules already installed in the rear of the chassis. You can install up to four additional 80 mm fan modules for a total of eight 80 mm fan modules.

See [“Installing components” on page 41](#) to determine the number of 80 mm fan modules that are required and where they should be installed in your chassis configuration.

To install a 80 mm fan module, complete the following steps.



Step 1. Remove the fan module filler, if one is installed.

Step 2. Grasp the fan module by the handle and align it with the fan bay.

Step 3. Slide the fan module into the chassis until it locks in place.

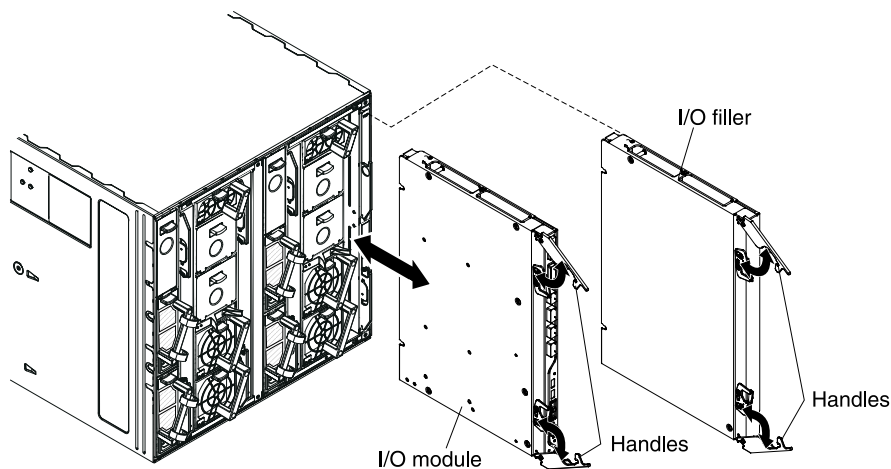
Installing an I/O module

You can install up to four I/O modules into the Flex System Enterprise Chassis. An Ethernet switch or a pass-thru module must be installed in I/O bay 1 and/or bay 2 whenever one or more compute nodes on-board Ethernet interface is active or there is an Ethernet I/O expansion card interfacing with I/O bay 1 and 2.

Before installing an I/O module, complete the following steps:

1. Verify that the I/O module is compatible with the chassis. See <http://www.lenovo.com/serverproven/>.
2. Read the installation instructions that come with the I/O module.

To install an I/O module, complete the following steps.



- Step 1. Remove the I/O filler, if necessary. Open the release handles (rotate the top handle up and the bottom handle down).
- Step 2. Slide the filler out of the bay.
- Step 3. Open the release handles on the I/O module (rotate the top handle up and the bottom handle down).
- Step 4. Align the I/O module with the bay on the chassis and slide the module into the module bay until it is seated.
- Step 5. Close the release handles (rotate the top handle down and bottom handle up).

Network integration with the Flex System Manager

In Flex System Enterprise Chassis with the Flex System Manager management node, you have the option to configure separate management and data networks.

The network configuration is established by the Flex System Manager management software during initial setup. You can also change the network configuration after initial setup by clicking the **Administration** tab in the Flex System Manager user interface.

You can use Flex System Manager management software to configure the network attributes for each network. Configuration parameters such as IPv4 or IPv6 address, subnet mask (IPv4), domain name servers, gateways, DHCP, and customized network routing information for each network are supported.

Important: When you establish two separate networks, establish network routing entries that define which network should be used for various network segments within the larger network.

Note:

- The illustrations in this section show management and data network examples when a single Chassis Management Module is installed in the chassis. Dual CMMs provide redundancy for the management fabric within the chassis as well as two separate uplinks from the chassis to the external network devices. You may choose to connect the redundant CMMs to redundant external Ethernet fabrics for added fault protection. The redundant CMMs within a chassis will automatically select the correct mode of operation based upon chassis and network status.
- Due to the design of the Flex System Enterprise Chassis, it is important to have all management network traffic on a separate subnet than the data traffic. This means all CMMs, IMMs, FSPs, and switch management interfaces should be on a different subnet/VLAN than the data production traffic of the other chassis components (such as node operating systems).

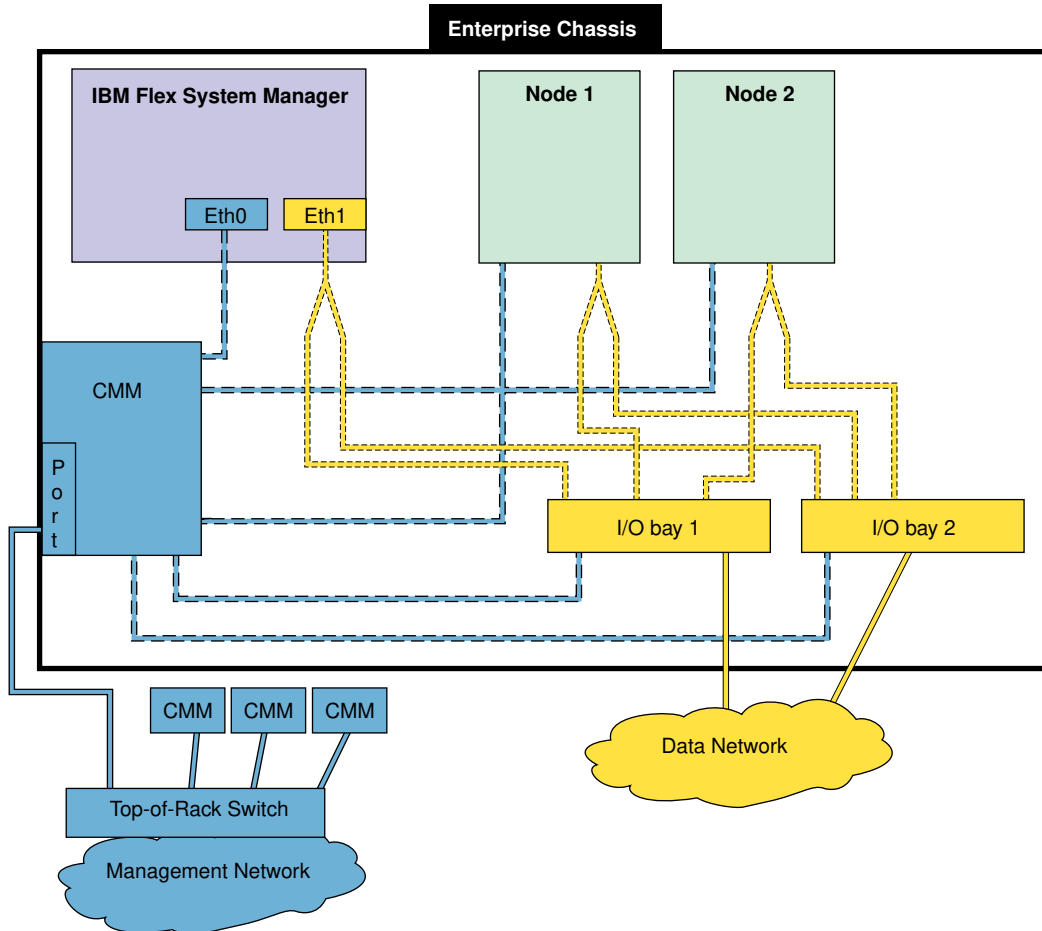
Management network

The Flex System Manager management node has two Ethernet interfaces which can be configured. The Eth0 port is connected to the Chassis Management Module (CMM) external Ethernet port through the chassis internal management network, effectively extending the chassis management network outside of the chassis. The CMM is connected to a top-of-rack switch that provides a central connection point for all chassis hardware to be managed by the Flex System Manager management software. These connections are referred to as the *management network*.

The management network is used to complete management-related functions for the various endpoints that are managed by the Flex System Manager management software, such as other Flex System Enterprise Chassis and compute nodes. During initialization the management software discovers any Flex System Enterprise Chassis on the management network. Note that the management node console can be connected to the management network or to the data network.

Note: If you want to configure your network so that the entire network is set up on a single IP subnet, configure just Eth0 on the Flex System Manager management node. If you choose to configure Eth0 and Eth1, you must configure them on different IP subnets.

Example of Separate Management and Data Networks



Data network

The Flex System Manager management node Eth1 port must be connected to the chassis switch modules that are installed in I/O bay 1 or bay 2. This is referred to as the *data network*. You can configure a switch module in bay 1 or bay 2 to map Eth1 to one of its external Ethernet ports, as you would configure the other nodes in the chassis that are connected to the external network. The data network is used by applications and operating systems and can support data transfer rates up to 10 Gbps if a chassis switch module that is capable of 10 Gbps is installed.

One of the key functions that the data network supports is discovery of operating systems on the various network endpoints. Discovery of operating systems by the Flex System Manager is required to support software updates on an endpoint such as a compute node. The Flex System Manager Checking and Updating Compute Nodes wizard assists you in discovering operating systems as part of the initial setup.

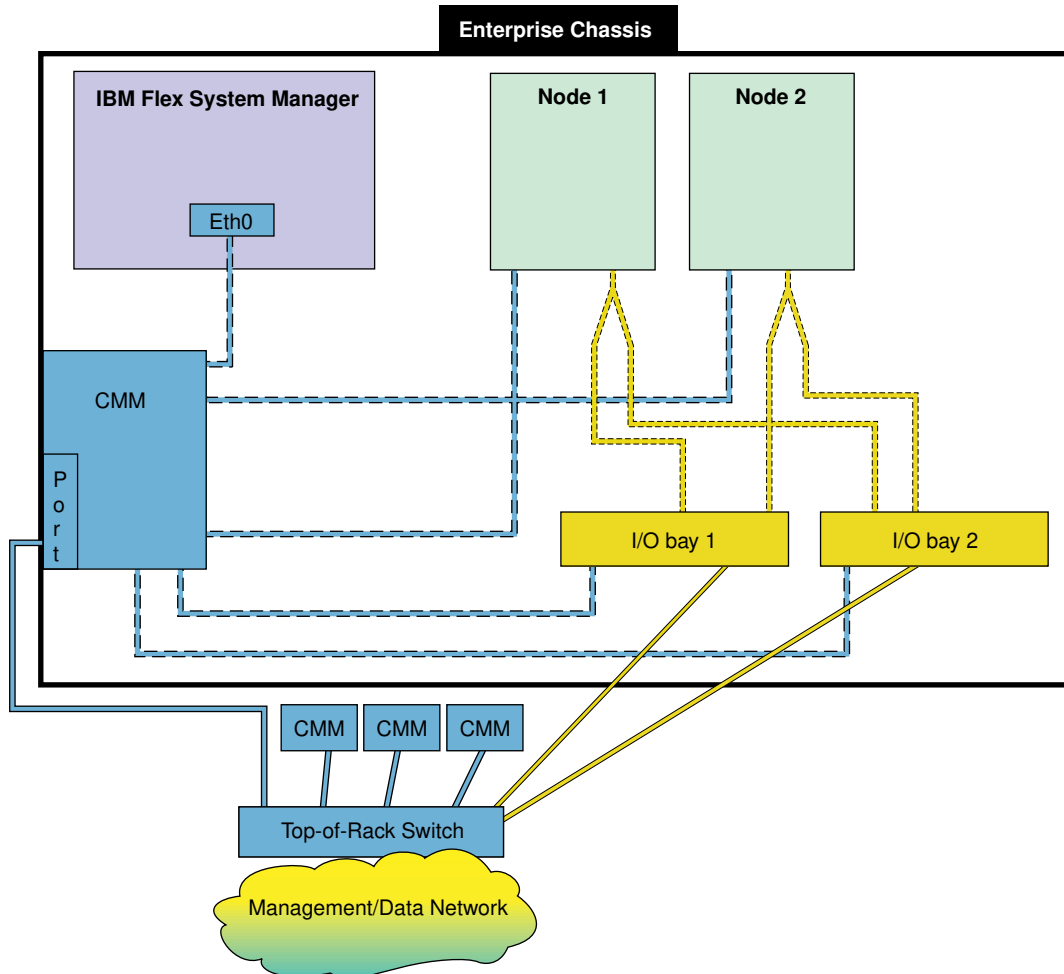
When you set up a high-availability environment that consists of active and passive Flex System Manager management nodes, either the management network or the data network can be used to replicate the data from the active management node to the passive management node. The additional bandwidth that is supported on the data network can improve data replication performance between active and passive management nodes.

Single network configuration

You can also set up your environment so that the management network and data network are the same network. This is typically a simpler approach to network configuration and results in an easier setup process. Typically, a single gateway can be used, and network routing table configuration can use default values that are established by the Flex System Manager setup wizard.

Note: When you are using a single network configuration, users and applications that have access to the data network also have access to the management network. If separating data and management access is important for security considerations, you should use separate data and management networks.

Example of Single Management and Data Network



Cabling the chassis

Cabling the Flex System Enterprise Chassis requires a site integration plan that describes the network environment. Make sure that you have completed all planning activities before you connect the network cables.

Cabling to the network components in the Flex System Enterprise Chassis will vary depending on the number of chassis, the type of I/O modules and management devices that you have purchased, and the planned network environment. See the site integration plan for more information about cabling the network.

Planning information is available from <http://flexsystem.lenovofiles.com/help/index.jsp>.

Information about cabling network switches is available from http://flexsystem.lenovofiles.com/help/topic/com.lenovo.acc.networkdevices.doc/network_iomodule.html.

See “[Connecting the chassis to power](#)” on page 55 for information about connecting the chassis power cords.

Connecting the chassis to power

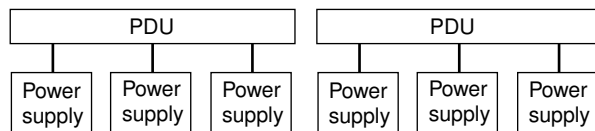
Power is supplied to the Flex System Enterprise Chassis when one end of each power cord is connected to a power connector on the rear of the chassis and the other end of each power cord is connected to a power distribution unit (PDU) or uninterruptible power source (UPS).

Connecting for N+N power redundancy

The Flex System Enterprise Chassis can have up to six power supplies. The chassis allows you to populate power supplies to meet the load demand that is installed in the chassis. As more nodes are installed in the chassis, you can install additional power supplies to meet the increased load demand.

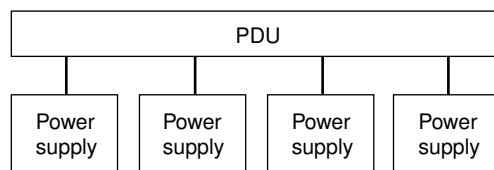
Attention: If there is not enough power available from the power supplies to meet the load demand installed in the chassis, the Chassis Management Module will not allow some of the devices to power on.

To provide power source redundancy for the chassis, you can connect the power supplies in an N+N configuration, where N can be 1, 2, or 3, depending on the total load installed in the chassis. In this configuration, the power-supply power cords are connected to separate PDUs. If a supply circuit fails, the remaining power supplies have enough power available to power the entire chassis load. A fully configured chassis with N+N power redundancy has six power supplies.



Connecting for N+1 power redundancy

If power source redundancy is not a concern but you want power supply redundancy, you can connect the power supplies in an N+1 configuration, where N can be 1, 2, 3, 4, or 5 depending on the total load that is installed in the chassis. In this configuration, the power-supply power cords are connected to the same PDU but there is one extra power supply available (+1). If one of the power supplies fails, the remaining power supplies have enough power available to power the entire chassis load. If two or more power supplies fail, it is possible for the entire chassis to lose power. If the power source circuit fails, the entire chassis will lose power.



Note: The Flex System Enterprise Chassis does not have a power switch. See “[Disconnecting the chassis from power](#)” on page 60 for more information.

Statement 29



CAUTION:

This equipment is designed to permit the connection of the earthed conductor of the dc supply circuit to the earthing conductor at the equipment. If this connection is made, all of the following conditions must be met:

- This equipment shall be connected directly to the dc supply system earthing electrode conductor or to a bonding jumper from an earthing terminal bar or bus to which the dc supply system earthing electrode conductor is connected.
- This equipment shall be located in the same immediate area (such as, adjacent cabinets) as any other equipment that has a connection between the earthed conductor of the same dc supply circuit and the earthing conductor, and also the point of earthing of the dc system. The dc system shall not be earthed elsewhere.
- The dc supply source shall be located within the same premises as this equipment.
- Switching or disconnecting devices shall not be in the earthed circuit conductor between the dc source and the point of connection of the earthing electrode conductor.

Statement 31

Electrical current from power, telephone, and communication cables is hazardous.
To avoid a shock hazard:

- Do not connect or disconnect any cables or perform installation, maintenance, or reconfiguration of this product during an electrical storm.
- Connect all power cords to a properly wired and grounded power source.
- Connect to properly wired power sources any equipment that will be attached to this product.
- When possible, use one hand only to connect or disconnect signal cables.
- Never turn on any equipment when there is evidence of fire, water, or structural damage.
- Disconnect the attached ac power cords, dc power sources, network connections, telecommunications systems, and serial cables before you open the device covers, unless you are instructed otherwise in the installation and configuration procedures.
- Connect and disconnect cables as described in the following table when you install, move, or open covers on this product or attached devices.

To Connect:

1. Turn OFF all power sources and equipment that is to be attached to this product.
2. Attach signal cables to the product.
3. Attach power cords to the product.
 - For ac systems, use appliance inlets.
 - For dc systems, ensure correct polarity of -48 V dc connections: RTN is + and -48 V dc is -. Earth ground should use a two-hole lug for safety.
4. Attach signal cables to other devices.
5. Connect power cords to their sources.
6. Turn ON all the power sources.

To Disconnect:

1. Turn OFF all power sources and equipment that is to be attached to this product.
 - For ac systems, remove all power cords from the chassis power receptacles or interrupt power at the ac power distribution unit.
 - For dc systems, disconnect dc power sources at the breaker panel or by turning off the power source. Then, remove the dc cables.
2. Remove the signal cables from the connectors.
3. Remove all cables from the devices.

Statement 34**CAUTION:**

To reduce the risk of electric shock or energy hazards:

- **This equipment must be installed by trained service personnel in a restricted-access location, as defined by the NEC and IEC 60950-1, First Edition, The Standard for Safety of Information Technology Equipment.**
- **Connect the equipment to a properly grounded safety extra low voltage (SELV) source. A SELV source is a secondary circuit that is designed so that normal and single fault conditions do not cause the voltages to exceed a safe level (60 V direct current).**
- **Incorporate a readily available approved and rated disconnect device in the field wiring.**
- **See the specifications in the product documentation for the required circuit-breaker rating for branch circuit overcurrent protection.**
- **Use copper wire conductors only. See the specifications in the product documentation for the required wire size.**
- **See the specifications in the product documentation for the required torque values for the wiring-terminal screws.**

Attention: The following circuit breaker and ground cable ratings apply to -48 V dc power supplies only:

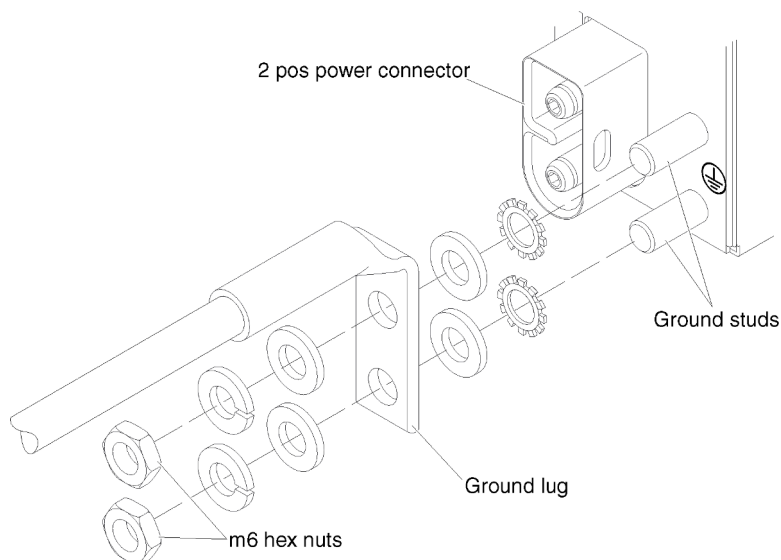
Breaker	Listed 70 A	See Note 1
Ground cable	4 AWG with Listed lug which can accept M6 ground screws	See Note 2
Torque rating for ground screws	4.0 - 4.8 Newton-meters (35.4 - 42.5 inch-pounds)	-
1. The maximum steady state current of the -48 V dc power supply is less than 70 A. However during specific events, such as over subscription, it is possible for the power supply to briefly draw a current greater than 70 A. Therefore it is recommended that the power supply be protected by a Listed circuit breaker that will support up to 90 A for a minimum of 20 ms. The suggested Telect High Current Panel Dual 350A Power Distribution Panel (part number 350CB06) using the Telect 70 A circuit breakers (Part number 090-0052-0070) conforms to this specification. 2. If not connecting to a SELV source which provides Reinforced insulation you must use a Ground Cable.		

To connect the chassis to power, complete the following steps:

1. For an ac-powered chassis (**restricted-access location is not required**), connect each power cord from the power supplies to a power distribution unit (PDU), uninterruptible power source (UPS) or wall receptacle. AC power is supplied to the Flex System Enterprise Chassis by one of the following options:
 - a. A power cord that connects to a PDU or UPS supplying a maximum 20 A branch circuit protection.
 - b. A power cord that connects to a wall receptacle supplying a maximum 20 A branch circuit protection.
 - c. A special use Flex System 3X Power Cord that connects to a wall receptacle supplying a maximum 32 A branch circuit protection.

Attention: Do not route the power cords over removable modules or allow the cords to interfere with the module handles.

2. For dc-powered chassis:
 - a. For a chassis powered by -48 to -60 V dc power supplies (**restricted access location is required**), connect an earth ground cable to each power supply.
 - 1) Use a 10 mm nut driver to remove the hex nuts from the ground studs.
 - 2) Remove the lock washer and one of the flat washers from each ground stud.
 - 3) Push the ground lug onto the ground studs; then, place the flat washer, the lock washer, and the hex nut back on each ground stud.
 - 4) Use a 10 mm nut driver to tighten the hex nuts to 4.0 - 4.8 Newton-meters (35.4 - 42.5 inch-pounds).



- b. Connect each power cord from the dc power supplies to a dc power distribution unit (PDU) (**restricted-access location is not required**).

Attention:

- Do not route the power cords over removable modules or allow the cords to interfere with the module handles.
- In North America, connect the power cords to a UL-listed PDU only.

3. Make sure that the following LEDs are lit:
 - The logo on the chassis front information panel.
 - The dc power and ac power LEDs on each power supply.
 - The power LED on each I/O module.

Note: The power LED on each compute node and on the Flex System Manager management node, if one is installed, flashes slowly to indicate that the node is connected to power and is ready to be turned on.

4. If the LEDs are not lit:
 - a. Disconnect the chassis from the power source.
 - b. Reseat all of the components in the chassis.
 - c. Reconnect the chassis to the power source.
 - d. If the problem persists, contact Support.

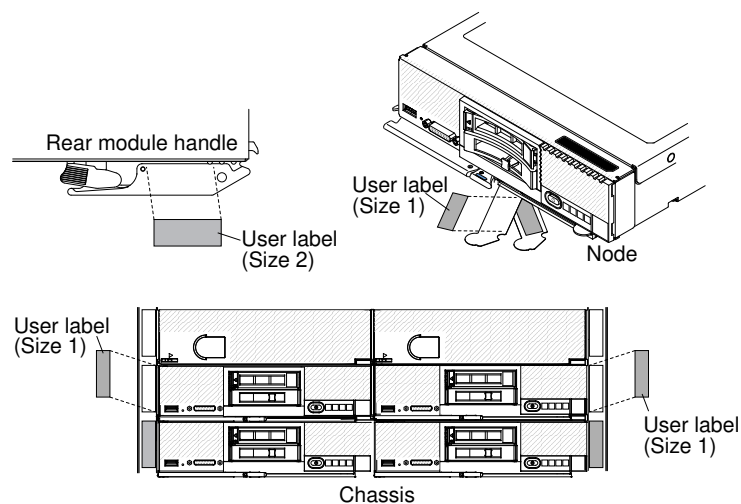
User labels

This section describes the user labels that come with the chassis and shows where to place them.

A set of blank user labels comes with the Flex System Enterprise Chassis. Record the information to identify each installed compute node, storage node, or Flex System Manager management node on the labels and place them on the node label tab and on the adjacent chassis label plate. The large label (size 1) fits on the node label tabs and on the chassis label plates.

Important: Do not place the label directly on the compute node or Flex System Manager management node front bezel or in any way block the ventilation holes.

The small label (size 2) fits on the rear chassis module handles. Use the small labels to record identifying information for the I/O modules and the CMM.



Obtaining firmware updates

You can use the Chassis Management Module or a management node, if installed, to update the chassis firmware.

The following chassis components have firmware that can be updated:

- Chassis Management Module
- Compute nodes
- I/O modules
- Flex System Manager management node

In addition, some of the optional hardware devices that you can order for the chassis have device drivers that you must install.

For example, Ethernet controllers are integrated on each compute node system board. The Ethernet controllers provide 1000 Mbps full-duplex capability only, which enables simultaneous transmission and reception of data to the external ports on the Ethernet switches. You do not have to set any jumpers or configure the controller for the compute node operating system. However, you must install a device driver in the compute node to enable the compute node operating system to address the Ethernet controller.

See the documentation that comes with your optional hardware devices for information about installing any required device drivers.

Attention: Before you install any new components in the Flex System Enterprise Chassis, update the Chassis Management Module firmware to the latest level available. See [Updating the CMM firmware](#) "Updating the CMM firmware" in the *Lenovo Flex System Chassis Management Module Installation Guide* for more information.

Using the Chassis Management Module

For more information about using the Chassis Management Module web interface to update the chassis firmware, see [Lenovo Flex System Chassis Management Module Chassis management options](#) "Chassis management options" in the *Lenovo Flex System Chassis Management Module User's Guide*.

You can use the Chassis Management Module command-line interface to update the chassis firmware. See [Lenovo Flex System Chassis Management Module update command](#) "update command" in the *Lenovo Flex System Chassis Management Module Command-Line Interface Reference Guide* for more information.

Using a management node

If a Lenovo XClarity Administrator is available, you can download, install, and manage firmware updates for managed endpoints, including chassis, compute nodes, and I/O modules. You can assign compliance policies to the managed endpoints to ensure that firmware on those endpoints remains compliant.

Note: Firmware updates can be applied to the hardware only. You cannot update device drivers using the Lenovo XClarity Administrator.

For more information about using the Lenovo XClarity Administrator, see http://flexsystem.lenovofiles.com/help/topic/com.lenovo.lxca.doc/aug_product_page.html.

You can also use the Flex System Manager management node to update firmware and device drivers for all of the chassis components, if one is available. For more information about using the Flex System Manager management software to update firmware and device drivers for the chassis components, see <http://flexsystem.lenovofiles.com/help/index.jsp>.

Disconnecting the chassis from power

Follow the instructions in this section to disconnect the Flex System Enterprise Chassis from all power sources.

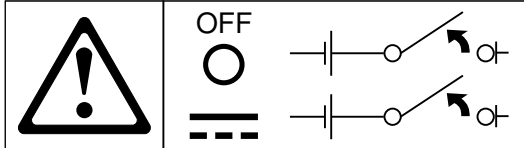
Before you disconnect the Flex System Enterprise Chassis from power, be sure to follow these instructions to correctly shut down the compute nodes.

Statement 19



CAUTION:

The power-control button on the device does not turn off the electrical current supplied to the device. The device also might have more than one connection to dc power. To remove all electrical current from the device, ensure that all connections to dc power are disconnected at the dc power input terminals.

**Statement 31**

Electrical current from power, telephone, and communication cables is hazardous.

To avoid a shock hazard:

- Do not connect or disconnect any cables or perform installation, maintenance, or reconfiguration of this product during an electrical storm.
- Connect all power cords to a properly wired and grounded power source.
- Connect to properly wired power sources any equipment that will be attached to this product.
- When possible, use one hand only to connect or disconnect signal cables.
- Never turn on any equipment when there is evidence of fire, water, or structural damage.
- Disconnect the attached ac power cords, dc power sources, network connections, telecommunications systems, and serial cables before you open the device covers, unless you are instructed otherwise in the installation and configuration procedures.
- Connect and disconnect cables as described in the following table when you install, move, or open covers on this product or attached devices.

To Connect:

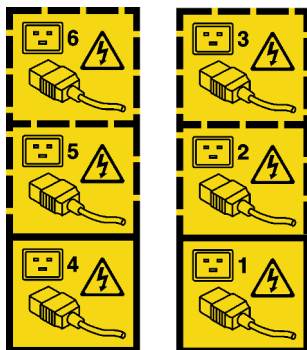
1. Turn OFF all power sources and equipment that is to be attached to this product.
2. Attach signal cables to the product.
3. Attach power cords to the product.
 - For ac systems, use appliance inlets.
 - For dc systems, ensure correct polarity of -48 V dc connections: RTN is + and -48 V dc is -. Earth ground should use a two-hole lug for safety.
4. Attach signal cables to other devices.
5. Connect power cords to their sources.
6. Turn ON all the power sources.

To Disconnect:

1. Turn OFF all power sources and equipment that is to be attached to this product.
 - For ac systems, remove all power cords from the chassis power receptacles or interrupt power at the ac power distribution unit.
 - For dc systems, disconnect dc power sources at the breaker panel or by turning off the power source. Then, remove the dc cables.
2. Remove the signal cables from the connectors.
3. Remove all cables from the devices.

Statement 33**CAUTION:**

This device does not provide a power control button. Removing power supply modules or turning off the server blades does not turn off the electrical current supplied to the device. The device also might have more than one power cord. To remove all electrical current from the device, ensure that all power cords are disconnected from the power source.



Attention: The following circuit breaker and ground cable ratings apply to -48 V dc power supplies only:

Breaker	Listed 70 A	See Note 1
Ground cable	4 AWG with Listed lug which can accept M6 ground screws	See Note 2
Torque rating for ground screws	4.0 - 4.8 Newton-meters (35.4 - 42.5 inch-pounds)	-
1. The maximum steady state current of the -48 V dc power supply is less than 70 A. However during specific events, such as over subscription, it is possible for the power supply to briefly draw a current greater than 70 A. Therefore it is recommended that the power supply be protected by a Listed circuit breaker that will support up to 90 A for a minimum of 20 ms. The suggested Telect High Current Panel Dual 350A Power Distribution Panel (part number 350CB06) using the Telect 70 A circuit breakers (Part number 090-0052-0070) conforms to this specification. 2. If not connecting to a SELV source which provides Reinforced insulation you must use a Ground Cable.		

To disconnect the chassis from power, complete the following steps:

- Step 1. Shut down each compute node. See the documentation that comes with the compute nodes for information about shutting down the operating systems on the compute nodes.
- Step 2. Shut down each storage node. See the documentation that comes with the storage node for information about shutting down the node.
- Step 3. Shut down the Flex System Manager management node, if one is installed.

Important: If the management node is the active management node in a high-availability configuration (two management nodes), perform a manual failover to the passive management node before you shut down the active management node. See the documentation that comes with the management node for information about high availability and shutting down the management node.

- Step 4. Disconnect all power cords on the rear of the chassis from the power source.

Note: After you disconnect the chassis from power, wait at least 5 seconds before you connect the chassis to power again.